



Rambler

AI Video Captioning Agent • Multi-Model Gemma Cascade

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POWERED BY GOOGLE GEMMA & AMD ROCm

01. The Problem & Solution

■ The Problem: Flat & Static Captions

- Traditional captions are dry, static, and context-blind.
- Static language models lack timing and cannot grasp comedic nuances, sarcasm, or tech humor.
- Standard video tools lack validation, leaving hallucinations and typos in output.
- Cloud API requirements compromise data privacy and lead to unpredictable network latencies.

■ The Solution: Multi-Agent Cascade

- Rambler uses specialized local agents for Ingestion, Multimodal Perception, and Styling.
- Automatically generates 4 tailored voices: Formal, Sarcastic, Tech Humor, and Fun/Everyday.
- Runs a Gemma-powered self-critique/eval loop that automatically refines low-scoring lines.
- Runs entirely on-device via AMD ROCm and vLLM for high performance and complete privacy.

02. Technical Architecture



03. Technology Stack & AMD ROCm

■ Gemma 4 Models

- google/gemma-4-31b-it
Handles vision + audio
frame-by-frame analysis.
- google/gemma-4-26b-a4b-it
Writes captions and handles
evaluation grading loops.

■ ROCm & vLLM

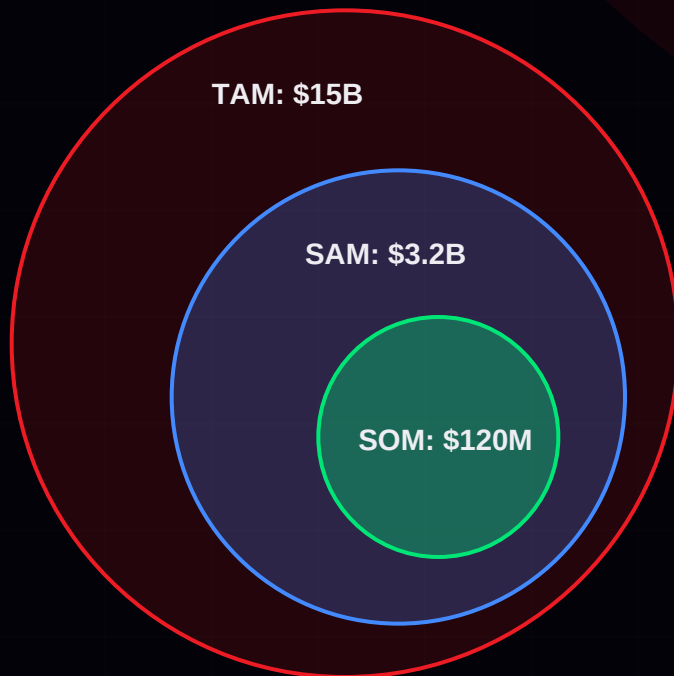
- AMD ROCm Accelerators
High-bandwidth HBM3 kernel
optimization.
- vLLM Model Servers
Serves endpoints on host
ports 8080 and 8081.

■ Docker Runtime

- Submission Dockerfile
Auto-detects batch task mode
or Streamlit web view.
- --network=host Bind
Shares ports directly with
vLLM host servers.

```
$ docker run --network=host -e INFERENCE_BACKEND=local -v ${PWD}/input:/input -v ${PWD}/output:/output rambler:latest
```

04. Market Potential & Revenue



Market Opportunity (TAM/SAM/SOM)

- Total Addressable Market (TAM): \$15B Creator economy and Video AI SaaS
- Serviceable Addressable Market (SAM): \$3.2B B2B localization/caching APIs
- Serviceable Obtainable Market (SOM): \$120M Tech & developer channels

Revenue Model & Projected Splits:

SaaS Subscriptions (45%)		\$15-\$49/mo
Enterprise API (35%)		\$0.02/min
Brand Tuning Packs (20%)		Custom LoRA

05. Competitive Advantage

Feature	Standard Auto-Cap	Whisper Speech	Claude Text-Only	RAMBLER
Multimodal Fusion (A+V)	☐ Visual Only	☐ Audio Only	☐ No Video	☒ Integrated (A+V)
Diverse Brand Styles	☐ Single Style	☐ Transcribe Only	☒ Can Style	☒ 4 Diverse Styles
Evaluation & Correction	☐ No Evaluation	☐ No Correction	☐ Manual	☒ Auto-Self-Correction
Privacy & Local ROCm	☐ Public Cloud Only	☒ Local Option	☐ Public Cloud Only	☒ Fully Local ROCm vLLM

06. Future Outlook & Scaling

1. Real-Time Streaming

- Live Broadcast Captions
Stream segment processing
for sub-second live feeds.
- Audio Sync Anchoring
Automatic timestamp alignment
using Whisper timestamps.

2. Custom Voice Adaption

- LoRA Adapters
Brands upload lightweight
fine-tuned adapters for
highly bespoke humor styles.
- Collaborative Editor
Manual correction triggers
reinforcement learning.

3. Scaled GPU Clusters

- Ray & K8s Clusters
Distribute video tasks
across multiples MI300X nodes.
- FP8 Quantization
Leverage ROCm FP8 tensor
kernels to double
inference throughput.